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PUMP HAVING A FLANGE FOR MOUNTING AN AUXILIARY PUMP

Abstract

A pump is disclosed. The pump may have a housing connectable to a transmission of a machine adjacent a front end of the pump. The pump may also have a flange attached to the housing adjacent a rear end of the pump. The flange may have a longitudinal axis extending along a length of the flange, and a transverse axis extending along a width of the flange. The flange may also have a first mounting hole configured to connect an auxiliary pump to the flange. Further, the flange may have a second mounting hole configured to connect a bracket to the flange. The first mounting hole may be positioned on a first axis inclined at an acute angle relative to the longitudinal axis of the flange.

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Background/Summary

CLAIM OF PRIORITY [0001] This application is a continuation of U.S. patent application Ser. No. 17/736,247, filed on May 4, 2022, the benefit of priority of which is claimed hereby, and which is incorporated by reference herein in its entirety.

TECHNICAL FIELD

[0002] The present disclosure relates generally to a pump, and more particularly, to a pump having a flange for mounting an auxiliary pump.

BACKGROUND

[0003] An industrial machine, for example, wheel loader, excavator, shovel, continuous miner, loader, or truck, includes an engine that provides power for propelling the machine and for operation of one or more work tools of the machine. Operation of the engine and/or the machine may require a variety of fluids. For example, the engine requires fuel for combustion. The engine also requires engine lubricant for lubrication of its moving parts, and coolant to cool various engine components. Likewise the machine may require transmission fluid to lubricate components of a transmission connecting the engine with wheels or tracked undercarriages of the machine, brake fluid to activate brakes on the wheels, and/or hydraulic fluid for the operation of one or more work implements.

[0004] The fluids required for operation of the machine may be supplied by one or more pumps that may be driven by the engine. In some configurations, at least some of the pumps of the machine may be assembled in the form of a pump stack with the pumps attached to each other back-to-back in a tandem configuration. In such a configuration, the first pump in the pump stack may be connected to and driven via a transmission or gear box associated with the engine, while the other pumps in the pumps stack may be driven by the shaft of the first pump. In some configurations, it may be necessary to add an additional pump to the pump stack to supply additional fluid to an existing component of the machine or to supply fluid to a new component being added to the machine. For example, an additional working implement may be added to the machine for performing operations at a worksite. Addition of one or more pumps to the pump stack may increase a weight of the pump stack and may require additional structural elements for mounting the pump to the chassis or to the transmission of the machine. Addition of one or more pumps to the pump stack may also require modifications to or replacement of one or more pumps already present in the pump stack.

[0005] U.S. Pat. No. 8,807,972 B2 of Reighard et al. issued on Aug. 19, 2014 (“the '972 patent”) and discloses a pump assembly for connecting a plurality of pumps in stack relation to one another. The '972 patent discloses a pump stack having three pumps connected to each other back-two-back in a tandem configuration. The pump stack of the '972 patent includes a first pump connectable to a gear configured to drive the pumps. Further, the pump stack includes retaining rods for connecting a second pump to a rear of the first pump, and for connecting a third pump to a rear of the second pump. In particular, the '972 patent discloses a first retaining rod that extends through the first and second pumps, and a second retaining rod that extends through the second and the third pump. The retaining rods are spring-loaded on one end and include a retaining clip to connect an opposite end of the rod to a respective pump housing.

[0006] Although the '972 patent discloses an assembly for attaching pumps to each other in a back-two-back tandem configuration, the pump stack assembly of the '972 patent may be sub-optimal. For example, addition of a pump to the pump stack of the '972 patent may require machining the new pump to allow for assembly of retaining rods. Moreover, the existing pumps in the pump stack

may have to be replaced or machined with additional holes to accommodate retaining rods for assembly of the new pump. Furthermore, the use of springs and retaining clips for assembly of the pumps may allow relative lateral movement of adjacent pumps in the pump stack of the '972 patent due to vibrations of the machine. Such relative movement may exert additional forces on the bearings of the shafts driving the pumps, potentially decreasing the reliability of the bearings and the pumps. Moreover, the retaining rods of the '972 patent are not attached to the gearbox and do not support the weight of the pumps in the pump stack. Thus, the entire weight of the pump stack is supported by the fasteners connecting the first pump in the stack to the gearbox. Such a configuration may induce additional stresses on the gearbox and/or the first pump due to the weight of the pumps in the pump stack, which in turn may decrease the reliability of the bearings and pumps in the pump stack.

[0007] The pump and flange of the present disclosure solve one or more of the problems set forth above and/or other problems of the prior art.

SUMMARY

[0008] In one aspect, the present disclosure is directed to a pump. The pump may include a housing connectable to a transmission of a machine adjacent a front end of the pump. The pump may also include a flange attached to the housing adjacent a rear end of the pump. The flange may include a longitudinal axis extending along a length of the flange, and a transverse axis extending along a width of the flange. The flange may also include a first mounting hole configured to connect an auxiliary pump to the flange. Further, the flange may include a second mounting hole configured to connect a bracket to the flange. The first mounting hole may be positioned on a first axis inclined at an acute angle relative to the longitudinal axis of the flange.

[0009] In another aspect, the present disclosure is directed to a pump stack for a machine. The pump stack may include a first pump mounted at a front end to a gearbox of the machine. The pump stack may also include a second pump mounted to a rear end of the first pump. A housing of the second pump may include a flange. The flange may include a longitudinal axis extending along a length of the flange, and a transverse axis extending along a width of the flange. The flange may also include a first pair of mounting holes configured to connect an auxiliary pump to the housing. In addition, the flange may include a second pair of mounting holes configured to connect a bracket to the housing. The first pair of mounting holes may be positioned on a first axis inclined at an acute angle relative to the longitudinal axis of the flange.

[0010] In yet another aspect the present disclosure is directed to a machine. The machine may include a chassis supported by a plurality of wheels and at least one working implement connectable to the chassis. The machine may also include an engine and a gearbox driven by the engine. The gearbox may be connectable to the engine and the wheels. The machine may include a pump stack driven by the engine. The pump stack may be configured to supply hydraulic fluid to the at least one working implement. The pump stack may include a first pump having a front end mounted to the gearbox. The pump stack may also include a second pump mounted to a rear end of the first pump. A housing of the second pump may include a flange. The flange may include a longitudinal axis extending along a length of the flange, and a transverse axis extending along a width of the flange. The flange may also include a first pair of mounting holes configured to connect an auxiliary pump to the housing and a second pair of mounting holes. The first pair of mounting holes may be positioned on a first axis inclined at an acute angle relative to the longitudinal axis of the flange. The pump stack may include a pump shaft configured to be driven by the engine. The pump shaft may include a splined recess configured to engage with a shaft of the auxiliary pump. The machine may also include a bracket having one end connected to the pump stack via fasteners passing through the second pair of mounting holes. An opposite end of the bracket may be connected to at least one of the gearbox or the chassis of the machine.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0011] FIG. 1 is a pictorial illustration of an exemplary disclosed machine

[0012] FIG. 2 is an illustration of an exemplary disclosed pump stack assembly for the machine of FIG. 1;

[0013] FIG. 3 is a side view of the exemplary disclosed pump stack assembly of FIG. 2, showing a pump stack having a pair of pumps; and

[0014] FIG. 4 is an elevation view of an exemplary disclosed flange of a pump included in the pump stack of FIG. 3.

DETAILED DESCRIPTION

[0015] FIG. 1 illustrates an exemplary mobile machine **10**. In the depicted embodiment, the machine **10** is a wheel loader. It is contemplated, however, that machine **10** may embody another type of mobile machine such as an excavator, a shovel, a continuous miner, a loader, a truck, a track-type-tractor, a motor grader, an articulated haul truck, an off-highway mining truck, or another construction machine known in the art. Machine **10** may include chassis **12**, one or more traction devices **14** that support chassis **12** on a ground surface, at least one working implement **16**, and engine **18**. In one exemplary embodiment as illustrated in FIG. 1, traction devices **14** may take the form of wheels **14**, although other types of traction devices such as crawler tracks are also contemplated.

[0016] Engine **18** may be any suitable type of internal combustion engine, such as a gasoline, diesel, natural gas, or hybrid-powered engine. It is contemplated, however, that in some exemplary embodiments, engine **18** may be driven by electrical power. Engine **18** may be configured to propel the one or more traction devices **14** via transmission **20**. Engine **18** may also be configured to deliver power to operate one or more other components or accessory devices (e.g. pumps, fans, motors, generators, belt drives) associated with machine **10**. For example, engine **18** may be configured to drive one or more pumps that may be configured to deliver hydraulic fluid to hydraulic actuator **22**, which may be configured to move working implement **16**.

[0017] FIG. 2 illustrates an exemplary pump stack assembly **30**. Pump stack assembly **30** may include pump stack **32** attached to a portion of transmission **20** of machine **10**. In one exemplary embodiment as illustrated in FIG. 2, pump stack **32** may be attached to gearbox **34** that may be included in transmission **20**. Gearbox **34** and/or transmission **20** may in turn be attached to chassis **12** of machine **10**. Pump stack **32** may include one or more pumps that may be configured to pump fluids such as fuel, lubricant, coolant, and/or hydraulic fluid to various components of machine **10**. In one exemplary embodiment as illustrated in FIG. 2, pump stack **32** may include pumps **36**, **38**, and **40**. Although only three pumps **36**, **38**, and **40** are illustrated as making up pump stack **32**, it is contemplated that pump stack **32** may include more than or less than three pumps **36**, **38**, and **40**. Further, pumps **36**, **38**, and **40** may include a same type of pump (e.g., vane pump, piston pump, gear pump). Alternatively, some or all of pumps **36**, **38**, and/or **40** may be of different types from each other. Similarly, pumps **36**, **38**, and **40** may be configured to pump the same kind of fluid (e.g., hydraulic fluid) or may be configured to pump different kinds of fluids (e.g., fuel, coolant, lubricant).

[0018] First pump **36** may be attached to gearbox **34** and may include pump shaft **50** (see FIG. 3) that may engage with one or more gears within gearbox **34**. Second pump **38** may be attached to a rear end of first pump **36** and may include pump shaft **98** (see FIG. 4) that may engage with pump shaft **50**. Third pump **40** may be connected to rear end **48** of second pump **38** via one or more fasteners. Third pump **40** may also include a shaft that may engage with pump shaft **98** (see FIG. 4) of second pump **38**. Thus, engine **18** may be configured to drive first pump **36**, second pump **38**, and/or third pump **40** by causing the rotation of pump shaft **50** via gearbox **34**. It is contemplated,

that some exemplary embodiments of machine **10** may include only first pump **36** and second pump **38**. Other exemplary embodiments of machine **10** may include third pump **40**, which may be an auxiliary pump that may be added to pump stack **32** to provide additional fluid to one or more existing components of machine **10**. Alternatively, auxiliary pump **40** may be added to pump stack **32** to provide fluid to a new component or working implemented being added to machine **10**.

[0019] FIG. **3** illustrates a side view of pump stack **32**, including pumps **36** and **38**. As illustrated in FIG. **3**, first pump **36** may extend from front end **42** of pump **36** to rear end **44** of pump **36**. First pump **36** may be connected to gearbox **34** adjacent front end **42**. First pump **36** may include pump shaft **50** that may be configured to engage with one or more gears of gearbox **34** and may be driven by engine **18** via gearbox **34**. Second pump **38** may extend from front end **46** of pump **38** to rear end **48** of pump **38**. Front end **46** of pump **38** may be connected to rear end **44** of first pump **36** via one or more fasteners. Second pump **38** may include pump shaft **98** (see FIG. **4**) that may engage with pump shaft **50** of first pump **36**. Thus, rotation of pump shaft **50** by engine **18** may also drive second pump **38** by rotating pump shaft **98**.

[0020] As also illustrated in FIG. **3**, second pump **38** may include housing **54**. In one exemplary embodiment as illustrated in FIG. **3**, housing **54** may be a two-part housing, including front housing **56** and rear housing **58**. Front housing **56** may be connected to rear end **44** of pump **36**. Rear housing **58** may be connected to front housing **56**. Rear housing **58** of pump **38** may include flange **52**. Although second pump **38** has been illustrated in FIG. **3** as having a two-part housing, it is contemplated that in some exemplary embodiments, pump **38** may have a single housing **54**. In some exemplary embodiments, flange **52** may be integral with housing **54** in the form of a single casting. It is contemplated, however, that in other exemplary embodiments, flange **52** may be a separate piece that may be removably attached to housing **54** (or rear housing **58**) via fasteners. Alternatively, in some exemplary embodiments, flange **52** may be fixedly attached to housing **54** (or rear housing **58**) by welding, brazing, or other attachment methods.

[0021] Returning to FIG. **2**, third pump **40** (or auxiliary pump **40**) may be attached to flange **52** of rear end **48** of second pump **38** via fasteners **60**. As also illustrated in FIG. **2**, in some exemplary embodiments, pump stack assembly **30** may include bracket **62**. One end **64** of bracket **62** may be attached to flange **52** via fasteners **66**. Opposite end **68** of bracket **62** may be attached to gearbox **34**, another portion of transmission **20**, and/or to chassis **12** via fasteners **70**. Thus, a weight of pump stack **32** may be supported by gearbox **34** adjacent front end **42** of pump **36** and by bracket **62** adjacent rear end **48** of pump **38**. It is contemplated, however, that some exemplary embodiments of pump stack assembly **30** may not include bracket **62**.

[0022] FIG. **4** illustrates an elevation view of an exemplary disclosed flange **52** of pump **38** included in pump stack **32**. Flange **52** may be an elongated plate-like structure having two generally linear vertical edges **82** spaced apart from each other, and two generally linear horizontal edges **84** spaced apart from each other. Curvilinear edges **86** may connect vertical edge **82** with an adjacently located horizontal edge **84**. It is contemplated, however, that flange **52** may have other shapes, for example, rectangular, square, circular, oval, etc. In one exemplary embodiment as illustrated in FIG. **4**, flange **52** may have a length “L” that may be larger than a width “W.” It is contemplated, however, that in other exemplary embodiments, length L of flange **52** may be equal to or smaller than width W of flange **52**. As further illustrated in FIG. **4**, flange **52** may include longitudinal axis **90** extending along a length of flange **52**. Longitudinal axis **90** may be disposed generally parallel to vertical edges **82** of flange **52**. In one exemplary embodiment as illustrated in FIG. **4**, vertical edges **82** may be symmetrically positioned on opposite sides of longitudinal axis **90**. Flange **52** may also include transverse axis **92** extending along a width of flange **52**. Transverse axis **92** may be disposed generally parallel to a horizontal edges **84** of flange **52**. In one exemplary embodiment as illustrated in FIG. **4**, horizontal edges **84** may be symmetrically positioned on opposite sides of transverse axis **99**. Longitudinal axis **90** may be disposed generally perpendicular to transverse axis **92** of flange **52**. The term generally as used in this disclosure encompasses

typical machining and manufacturing tolerances. Thus, for example, generally perpendicular may encompass angles in the range $90^{\circ} \pm 1^{\circ}$, and generally parallel may encompass angles in the range of $0^{\circ} \pm 1^{\circ}$.

[0023] Flange **52** may include one or more mounting holes **94** configured to receive one or more fasteners **60** for attaching auxiliary pump **40** to flange **52**. In one exemplary embodiment as illustrated in FIG. 4, flange **52** may include a pair of mounting holes **94** positioned diametrically opposite from each other. Mounting holes **94** may be positioned at a radius “R.sub.1” relative to axis of rotation **96** (see FIG. 3) of pump shaft **98** of pump **38**. In one exemplary embodiment, mounting holes **94** may be positioned on opposite sides of longitudinal axis **90**, for example, on an axis **100** that may be inclined at an acute angle θ relative to longitudinal axis **90** of flange **52**. In one exemplary embodiment, axis **100** may pass through centers of mounting holes **94**. Flange **52** may also include one or more mounting holes **102** configured to receive one or more fasteners for attaching bracket **62** to flange **52**. Mounting holes **102** may be disposed at a radius “R.sub.2” relative to axis of rotation **96** of pump shaft **98** of pump **38**. In one exemplary embodiment as illustrated in FIG. 4, radius R.sub.1 may be smaller than radius R.sub.2. It is contemplated, however, that in other exemplary embodiments, radius R.sub.1 may be larger than or equal to radius R.sub.2. As also illustrated in the exemplary embodiment of FIG. 4, the pair of mounting holes **102** may be positioned on a pair of axes **104** inclined at an acute angle v relative to each other. In one exemplary embodiment, each axis **104** may pass through a center of an associated mounting hole **102**. In some exemplary embodiments as illustrated in FIG. 4, the pair of mounting holes **104** may be positioned on one side of longitudinal axis **90** and/or on one side of axis **100** passing through mounting holes **94**. As further illustrated in FIG. 4, at least one mounting hole **102** may be positioned on an axis inclined at an acute angle ϕ relative to transverse axis **92**. In one exemplary embodiment as illustrated in FIG. 4, at least one mounting hole **94** and at least one mounting hole **102** may be positioned on opposite sides of longitudinal axis **90**. In one exemplary embodiment, some or all of mounting holes **94** and/or **104** may be through holes. It is contemplated, however, that in other exemplary embodiments, some or all of mounting holes **94** and **104** may be threaded. Although particular locations of mounting holes **94** and **104** have been discussed above, it is contemplated that mounting holes **94** and **104** may be disposed at other positions on flange **52**.

[0024] Pump shaft **98** of pump **38** may engage with pump shaft **50** of pump **36** adjacent rear end **44** of pump **38**. Pump shaft **98** of pump **38** may include a generally cylindrical recess **106** extending into pump shaft **98** from adjacent rear end **48** of pump **38** towards front end **46** of pump **38**. In one exemplary embodiment as illustrated in FIG. 4, a wall of recess **106** may include a plurality of splines **108**. A shaft of auxiliary pump **40** may be configured to engage with splines **108** when auxiliary pump **40** is attached to flange **52**. Thus, pump shaft **50** of pump **36** may drive pump shaft **98** of pump **38**, which in turn may drive the shaft of auxiliary pump **40**.

INDUSTRIAL APPLICABILITY

[0025] The disclosed pump **38** with flange **52** may provide several advantages. For example, the arrangement and positions of mounting holes **94** on flange **52** may help ensure ease of assembly of an auxiliary pump **40** to pump **38**, particularly on an existing pump stack **32** during maintenance or repair of machine **10**. In some exemplary embodiments, an auxiliary pump **40** may be added to supply fluid when replacing an existing component on machine **10** with a component of larger size or when adding a new component to machine **10**. For example, an additional auxiliary pump **40** may be required to provide hydraulic fluid to a working implement added to machine **10** during maintenance or at a construction site. Positioning mounting holes **94** on axis **100** that is inclined relative to longitudinal axis **90** may allow for attachment of auxiliary pump **40** to pump **38** without the need for replacement, resizing, or machining of pump **38** and/or a need for changing a size of flange **52**. Further, use of removable fasteners **60** to attach auxiliary pump **40** may allow pump **38** and/or pump **40** to be easily connected to or disconnected from each other, for example, for

maintenance and/or repair. The use of fasteners **60** in mounting holes **94** may also allow pump **38** to be tightly fastened to pump **40** to help decrease the likelihood of lateral movement of pump **40** relative to pump **38**, minimizing the amount of radial forces that may be exerted on bearings supporting pump shaft **98** and/or a shaft of auxiliary pump **40**.

[0026] Additionally, mounting holes **102** on flange **52** may allow for attachment of bracket **62** to flange **52**. One end **64** of bracket **62** may be attached to flange **52** fasteners **66**. An opposite end **68** of bracket **62** may be attached to gearbox **34**, transmission **20** and/or chassis **12** via fasteners **70**. Thus, a weight of pump stack **32** may be supported by gearbox **34** adjacent front end **42** of pump **36** and by gearbox **34**, transmission **20** and/or chassis **12** adjacent rear end **48** of pump **38** via bracket **62**. Supporting pump stack **32** on both ends in this manner may minimize the effect of the weight of pump stack **32** on radial forces exerted on the fasteners attaching pump stack **32** machine **10** and/or on the bearings supporting pump shafts **50** and **98**. This in turn may help increase the reliability of pumps **36**, **38**, and **40** and of pump stack assembly **30**. Thus, the disclosed flange **52** and its arrangement of mounting holes **94** and **102** may help ensure ease of assembly/disassembly of auxiliary pump **40** to/from pump stack **32**, and may also minimize the risk of damage to shaft **50**, pump shaft **98**, and/or bearings supporting shaft **50** and/or pump shaft **98** due to a weight of pumps **36**, **38**, and **40**.

[0027] It will be apparent to those skilled in the art that various modifications and variations can be made to the disclosed pump and flange. Other embodiments will be apparent to those skilled in the art from consideration of the specification and practice of the disclosed pump and flange. It is intended that the specification and examples be considered as exemplary only, with a true scope being indicated by the following claims and their equivalents.

Claims

1. A pump stack for a machine, comprising: a first pump mounted at a front end to a gearbox of the machine, the first pump including a first pump shaft; a second pump mounted to a rear end of the first pump, the second pump including a housing, a flange protruding from the housing, and a second pump shaft configured to engage with the first pump shaft, the flange comprising: a longitudinal axis extending along a length of the flange and a transverse axis extending along a width of the flange, the longitudinal axis and the transverse axis together defining four quadrants of the flange; a first pair of mounting holes configured to connect an auxiliary pump to the housing, wherein the first pair of mounting holes are located at a first radial distance relative to the second pump shaft, and the first pair of mounting holes are positioned on a first axis inclined at a first acute angle relative to the longitudinal axis of the flange; and a second pair of mounting holes configured to connect a bracket to the housing, the second pair of mounting holes being located at a second radial distance relative to the second pump shaft, the second radial distance being greater than the first radial and the second pair of mounting holes both being located in a same quadrant of the four quadrants.
2. The pump stack of claim 1, wherein the second pair of mounting holes are positioned on respective axes inclined at a second acute angle relative to each other.
3. The pump stack of claim 2, wherein the respective axes are each inclined at respective third acute angles relative to the longitudinal axis of the flange.
4. The pump stack of claim 1, wherein the first pair of mounting holes are positioned on opposite sides of the longitudinal axis.
5. The pump stack of claim 1, wherein at least one mounting hole of the first pair of mounting holes and the second pair of mounting holes is threaded.
6. The pump stack of claim 1, wherein the second pump shaft includes a splined recess configured to receive a shaft of the auxiliary pump.
7. The pump stack of claim 1, wherein the bracket is attached to the housing of the second pump

via the second pair of mounting holes, the bracket also being attached to one of the gearbox or a chassis of the machine.

8. The pump stack of claim 1, wherein the bracket extends across a gap between the flange and one of the gearbox or a chassis of the machine, and the bracket further includes: a first end that is attached to the flange; and a second end that is attached to one of the gearbox or a chassis of the machine.

9. The pump stack of claim 8, wherein the bracket includes an elongated structural member extending from the first end to the second end.

10. The pump stack of claim 8, wherein the first end of the bracket is attached to at least one mounting hole of the second pair of mounting holes of the flange via a fastener, and the second end of the bracket is attached to the gearbox or the chassis via another fastener.

11. The pump stack of claim 10, wherein the first end of the bracket is attached to the second pair of mounting holes of the flange via a pair of fasteners.

12. The pump stack of claim 11, wherein the second end of the bracket is attached to the gearbox or the chassis via a pair of fasteners.

13. The pump stack of claim 1, wherein the flange includes a first flange surface connected to the housing of the second pump and a second flange surface disposed opposite to and separated from the first flange surface by a thickness of the flange.

14. The pump stack of claim 1, wherein the flange is integral with the housing of the second pump.

15. The pump stack of claim 1, wherein the flange is removably attachable to the housing of the second pump.

16. A pump stack for a machine, comprising: a first pump including a first front end and a first rear end opposite the first front end, the first front end of the first pump being mounted to a gearbox of the machine and extending along and spaced apart from the gearbox defining a gap between the gearbox and the pump stack; a second pump including a second front end and a second rear end opposite the second front end, the second front end of the second pump being mounted to the first rear end of the first pump, the second pump including a housing and including a flange protruding from the housing at the second rear end, the flange including: a first pair of mounting holes configured to connect a third pump to the housing of the second pump; and a second pair of mounting holes; and a bracket extending across the gap between the pump stack and the gearbox, the bracket having a first end connected to the flange via the second pair of mounting holes and a second end connected to the gearbox.

17. The pump stack of claim 16, wherein the bracket includes an elongated structural member extending from the first end to the second end.

18. The pump stack of claim 16, wherein the first pair of mounting holes are positioned on a first axis inclined at a first acute angle relative to the longitudinal axis of the flange.
